Chapter 5

Investment Decisions, Look Ahead and Reason Back

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Learning Objectives

- 5.1 Calculate discount rate to figure out the trade-off between current sacrifice and future gain.
- 5.2 Identify that discount rates are determined by the cost of capital, which is compared to returns to make decisions about investing in projects.
- 5.3 Describe the NPV rule and other shortcuts like pay-back periods used by companies to analyze investments.
- 5.4 Calculate break-even quantity and break-even prices and determine how these affect decisions to invest or shut down.
- 5.5 Describe how to choose contracts involving sunk cost

Chapter 5 – Main Points

- Investments imply willingness to trade dollars in the present for dollars in the future. Wealth-creating transactions occur when individuals with low discount rates (rate at which they value future vs current dollars) lend to those with high discount rates.
- Companies, like individuals, have different discount rates, determined by their cost of capital. They invest only in projects that earn a return higher than the cost of capital.
- The **NPV rule** states that if the present value of the net cash flow of a project is larger than zero, the project earns **economic profit** (i.e., the investment earns more than the cost of capital).
- Although NPV is the correct way to analyze investments, not all companies use it. Instead, they use break-even analysis because it is easier and more intuitive.
- **Break-even quantity** is equal to fixed cost divided by the contribution margin. If you expect to sell more than the break-even quantity, then your investment is profitable.
- **Avoidable costs** can be recovered by shutting down. If the benefits of shutting down (you recover your avoidable costs) are larger than the costs (you forgo revenue), then shut down. The break-even price is average avoidable cost.
- If you incur **sunk costs**, you are vulnerable to **post-investment hold-up**. Anticipate hold-up and choose contracts or organizational forms that minimize the costs of hold-up.
- Once relationship-specific investments are made, parties are locked into a trading relationship with each other and can be held up by their trading partners. Anticipate hold-up and choose organizational or contractual forms to give each party both the incentive to make relationship-specific investments and to trade after these investments are made.

Warm-Up

- Imagine your friend spent $1,000 on a non-refundable vacation. The forecast now says rain all week. They ask: 'Should I still go?' What would you say and why?

Compounding

- Determining whether to invest in a project, asset, *etc*. means we need to compare the future benefits to the future costs.
  - Discounting allows us to do this. But first, let's look at its opposite.
- Compounding is a way to determine the future stream of income from an investment today.
- To determine this, use the following formula:

$$\boldsymbol{Future\ value_t = present\ value\ x\ (1 + r)^t}$$

- Where r is the interest rate, and t is the time index.
- Present value is the value today of a future amount of money
- For example, you have $1,000 today, and can invest it at a constant interest rate of 5% for 5 years. Then that $1,000 today will generate $276.28 in five years or:

$$\$1{,}276.28 = \$1{,}000\ x\ (1 + 0.05)^5$$

- The rule of 72 tells us the number of years required to have our investment double
  - Simply divide 72 by the interest rate to determine this

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Compounding Example

- How much is \$10 worth five years from now at a 5% interest rate? 2% interest? 0% interest?

$$Future\ value_t = \$10\ x\ (1 + 0.05)^5 = \$11.27$$
$$Future\ value_t = \$10\ x\ (1 + 0.02)^5 = \$11.10$$
$$Future\ value_t = \$10\ x\ (1 + 0.00)^5 = \$10.00$$

- As the interest rate increases, the \$10 today generates more in the future. Think of this as an opportunity cost of money being saved as opposed to spent.
- Notice that as the interest rate decreases so to does the future value. At an interest rate of 0, present and future values are the same. Hence, you will loan money to someone with any positive interest rate.

Discounting

- Discounting is the opposite of compounding. Discounting determines the present value of a future investment or income stream.
- To determine this, use the following formula:

$$\boldsymbol{Present\ value} = \frac{\boldsymbol{Future\ Value}}{(\mathbf{1}+\boldsymbol{r})^{\boldsymbol{t}}}$$

- When firms sell assets/merge/*etc.* they will calculate the discounted value of the assets being sold.
- Discounting, in essence, helps investors, firms, *etc.* determine how much they need to save today in order to meet a predetermined amount in the future.
- Generally speaking, the present value of a future amount is less than the actual future amount.
- For example, suppose you are saving money for your son/daughter to go to college. You need \$80,000 to cover all expenses. Your son/daughter will begin college in 10 years. If you can invest your savings at 6% interest, how much do you need to save today?

$$\frac{\$80{,}000}{(1+0.06)^{10}} = \$44{,}671.58$$

Discounting Example

- What is the present value of $100 one year from now at an 10% discount (interest) rate?

$$Present\ value = \frac{\$100}{(1+0.10)} = \$90.91$$

- What is the present value of $100 one year from now at an 4% discount rate?

$$Present\ value = \frac{\$100}{(1+0.04)} = \$96.15$$

- Notice what the lower discount rate does. If the present value is the value today of some future amount, then the lower the discount rate the higher the value on future dollars. Stated differently, at higher discount rates, more value is placed on current dollars as more can be earned by investing current dollars.
- Consider the present value of $10,000 with a 20% discount rate over 25 years. This amounts to $104.83. The high discount rate places more value on current dollars (simply substitute in a low discount rate to see this).
- Consider building a professional football stadium miles from the city with the thought that within 20 years the city will have expanded and be near the stadium. But, 20 years being well off in the future means the present value gains today are very small.

A Slightly More Complex Discounting Example

Suppose you are wrongfully fired from your job and are suing for lost wages. You were earning $50,000/year. We also assume that you would have received a cost-of-living adjustment each year (say 2%). We also need to consider job termination other than wrongful firing (*e.g.* death). If you are 50 years old and at your current job you would retire at 65; what is the present value of your earnings if the interest rate is 5%?

$$\boldsymbol{Present\ value} = \frac{\boldsymbol{W(1+g)^t(1-m_t)}}{\boldsymbol{(1+r)^t}}$$

Where W is your wage, g is the growth rate, and m is the mortality rate (probability of dying) for different ages.

To solve this, you will need to use an actuarial table that uses similar data for yourself.

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A Slightly More Complex Discounting Example

Age	$W(1+g)^t$	$(1-m_t)$	$1/(1+r)^t$	$W(1+g)_t(1-m_t)/(1+r)_t$
50	$50,000.00	0.994844	1	$49,742.20
51	$51,000.00	0.994378	0.952381	$48,298.36
52	$52,020.00	0.993879	0.907029	$46,894.86
53	$53,060.40	0.993344	0.863838	$45,530.49
54	$54,121.61	0.992778	0.822702	$44,204.41
55	$55,204.04	0.992156	0.783526	$42,914.53
56	$56,308.12	0.991507	0.746215	$41,661.13
57	$57,434.28	0.990884	0.710681	$40,445.38
58	$58,582.97	0.99031	0.676839	$39,267.04
59	$59,754.63	0.989747	0.644609	$38,123.44
60	$60,949.72	0.989128	0.613913	$37,011.03
61	$62,168.72	0.988409	0.584679	$35,927.44
62	$63,412.09	0.987597	0.556837	$34,872.27
63	$64,680.33	0.986675	0.530321	$33,844.30
64	$65,973.94	0.98563	0.505068	$32,842.49
65	$67,293.42	0.984447	0.481017	$31,865.84
			Total	**$643,445.22**

Net Present Value

- How can we determine if an investment will be profitable?
  - **Net Present Value**
- Discounting a future stream of income payments less the cost, helps us determine if we should undertake an investment.
  - The key is to make sure you discount right.
  - To do so, a business will consider the opportunity cost of the interest rate involved. That means, they will look for other opportunities and base the interest rate they choose on the highest valued forgone interest rate.
- Suppose we are given the following:
  - The Freezer Burn company is deciding whether to invest in a new anti-freezer burn technology that needs an initial investment of $100,000. This will increase cash flows in the first year by $55,000 and $60,000 in the second year. The firm's current fixed costs are $20,000 and marginal cost is $100. The firm currently charges $150 per unit. If the interest rate is 8% in both years, what is the net present value of the cash flows? Will the firm undertake the investment at 8%?

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Freezer Burn Company

- New technology: \$100,000
- Income in year 1: \$55,000
- Income in year 2: \$60,000
- Fixed cost: \$20,000
- Marginal Cost: \$100
- Price: \$150
- Interest rate: 8%

$$\frac{\$55{,}000}{(1+0.08)^1} = \$50{,}925.93$$

$$\frac{\$60{,}000}{(1+0.08)^2} = \$51{,}440.33$$

\$50,925.93 + \$51,440.33 = \$102,366.30

Yes, the firm will undertake the investment at 8% since the income generated is greater than the cost of the technology (\$102,366.30 > \$100,000) and the firm earns \$2,366.30 from the investment.

Activity - NPV Decision Practice

Scenario:

A company considers a $200,000 investment in equipment. It expects cash inflows of $100,000 in Year 1 and $120,000 in Year 2. The cost of capital is 10%.

Questions:

1. What is the NPV of the project?
2. Should the company invest?

Activity - NPV Decision Practice

Given:

- Initial investment = $200,000
- Cash flows:
  - Year 1 = $100,000
  - Year 2 = $120,000
- Discount rate = 10%

- Step 1: Discount future cash flows
  - Year 1 PV = 100,000 / (1 + 0.10)^1 = 90,909
  - Year 2 PV = 120,000 / (1 + 0.10)^2 = 99,174
- Step 2: Calculate NPV
  - NPV = 90,909 + 99,174 − 200,000 = -9,917

- Do not invest.

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What Determines the Discount Rate?

The discount rate reflects the **opportunity cost of capital,** not just a bank interest rate.

A good discount rate accounts for:

- **Time value of money**
  - Receiving $1,000 today is worth more than receiving $1,000 five years from now, because today you can invest it, earn interest, or use it to avoid borrowing.
- **Alternative investment returns**
  - If you could invest your capital in a mutual fund earning 10%, then any business project must return more than 10% to justify the risk. That 10% becomes your opportunity cost.
- **Project-specific risk**
  - A startup software project in a new market is riskier than upgrading office printers. You might use a 15% discount rate for the software investment vs. 6% for the printer upgrade.
- **Inflation (for nominal rates)**
  - If expected inflation is 3% and your real return target is 5%, your nominal discount rate should be:
  - Nominal rate = 5% + 3% = 8%

Net Present Value and Nominal Interest Rates

- When selecting the interest rate, be careful to distinguish between the real interest rate and the nominal one
  - Real interest rate (***r***) – the interest rate before inflation (***i***) (the actual opportunity cost of borrowing)
  - Nominal interest rate (***n***) – the real interest rate adjusted for inflation

$$n = r + i$$

- For example, in the Freezer Burn problem the real interest rate was 8%. Suppose however that Freezer Burn was charged the nominal rate of 12% (meaning there is 4% inflation).
- The new NPV is:

$$\frac{\$55{,}000}{(1+0.12)^1} = \$49{,}107.14 \qquad \frac{\$60{,}000}{(1+0.12)^2} = \$47{,}831.63$$

$49,107.14 + $47,831.63 = **$96,938.78**

- And you should not invest in the technology.

Net Present Value vs Payback

Metric	NPV	Payback Period
Time Value of Money	Fully accounts for it	Ignores time value
Profitability Indicator	Considers all future cash flows	Stops at breakeven recovery
Strategic Use	Long-term decision making	Quick screening tool
Risk Handling	Can include risk-adjusted rate	Does not account for risk
Common Pitfall	Requires accurate cash flow forecasts	Ignores long-term gains

Example Scenario:

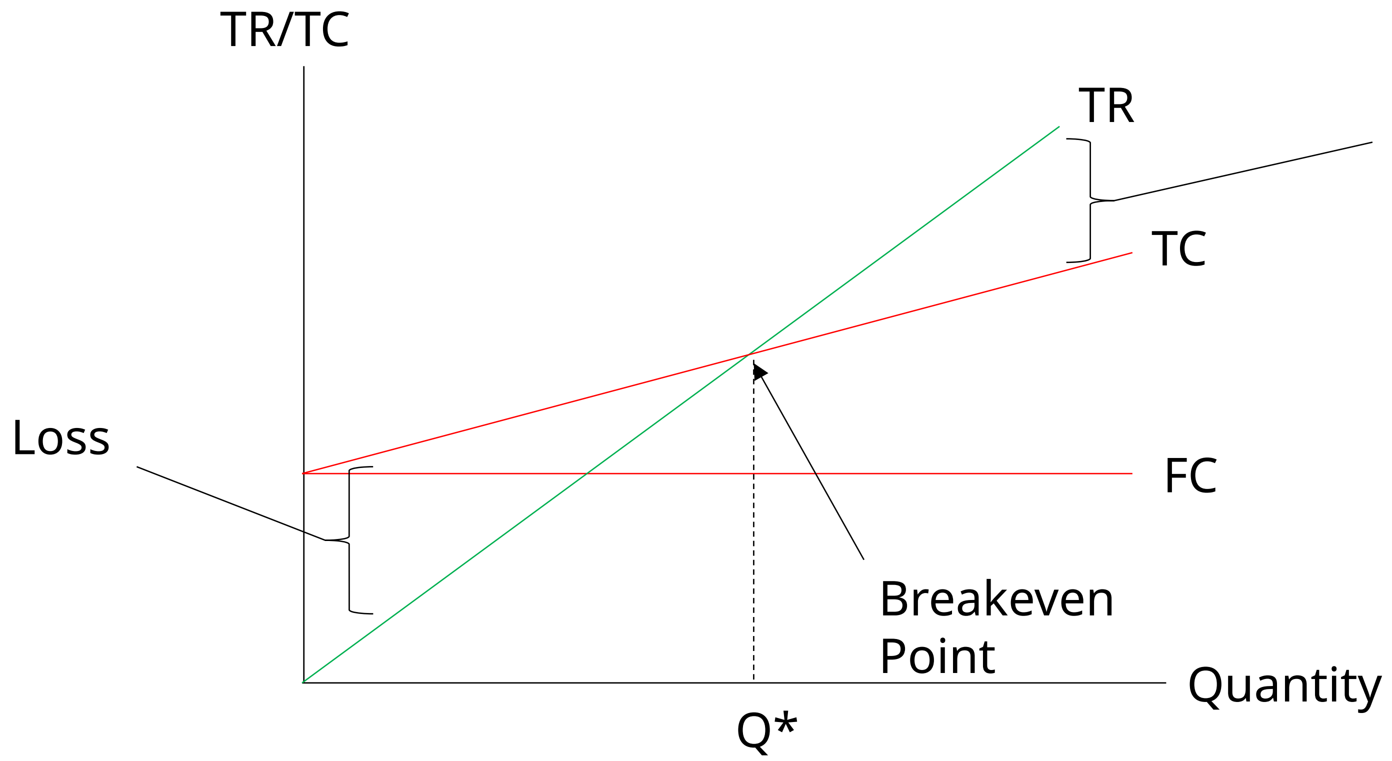
- Project A pays back in 2 years but has lower total returns.
- Project B takes 4 years to pay back but generates $50K more in NPV.

Discussion Activity 1

- Why might a company choose a project with a faster payback period over a project with a higher NPV?.
- Think about risk, liquidity needs, and managerial incentives. Are short-term wins sometimes more appealing?

Breakeven Analysis

- Breakeven analysis examines how much output it will take for your total revenue to just equal your total cost.
- By selling more than the breakeven quantity you earn the contribution margin, which leads to a profit. Keep in mind that fixed costs are also covered.
- Graphically:



Breakeven Analysis

$$Method\ 1$$

$$TR = TC$$

$$P * Q = FC + AVC * Q$$

$$(P - AVC) * Q = FC$$

$$Q = \frac{FC}{P - AVC}$$

$$Method\ 2$$

$$0 = TR - TC$$

$$0 = P * Q\ - (FC + MC * Q)$$

$$0 = (P - MC) * Q - FC$$

$$FC = (P - MC) * Q$$

$$\frac{FC}{P - MC} = Q$$

Breakeven Analysis

Fixed costs = $20,000
Cost of Capital = 10%
Variable costs = $4*Q*
Price = $14
What is the breakeven quantity?

AVC = $4

Contribution margin = $14 - $4 = $10

Breakeven quantity = (20,000 x 0.1) / 10 = 200

Breakeven Example

- A firm’s fixed costs are $3,500. The firm charges $50 for each unit. For every additional unit the firm produces, it costs the firm $15, and the cost of capital is 0%.

- What is the firm’s contribution margin?

Contribution margin = $50 - $15 = $35

- What is the firm’s breakeven quantity?

$$\frac{\$3{,}500}{\$50-\$15}=100$$

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Breakeven Example

- You remodeled a hotel in 2015 for $16 million. You currently forecast that you will have 10,000 hotel night stays that cost $25 per room per night to service. Your cost of capital is 15% (what you could have earned elsewhere). You can sell the hotel for $10 million. What is your breakeven price per room, per night?

$$\$10{,}000{,}000 \times 0.15 = \$1{,}500{,}000$$

$$\$1{,}500{,}000 / 10{,}000 = \$150$$

$$\$150 + \$25 = \$175$$

Discussion Activity 2

- If a proposal would increase sales but also fixed costs, what should you ask before saying yes?
- Try using the contribution rule: $(P - VC) \times \Delta Q > \Delta FC$. Does the extra revenue truly cover the cost?

Case Study: Lockheed’s Tri-Star Jet and Breakeven Analysis

- In 1971, Lockheed sought a guaranteed $250 million loan from the government.
- Lockheed calculated that it’s breakeven quantity would be 200 jets at $15.5 million each.
- They already had 103 jets ordered with an additional 75 in options.
- Lockheed received the loan but made a big mistake. It didn’t include the cost of developing the technology and construction facilities to build the jet into its fixed costs.
- Had Lockheed done so, the breakeven quantity would have been twice as high.
- Lockheed eventually scrapped it’s Tri-star airplane.
- Aircrafts usually require a large quantity before they breakeven. This is due to the large fixed costs from development and assembly.
- Consider Airbus. It took 20 years and $26 billion in government subsidies before Airbus reached its breakeven quantity in 1990.

Limitations of Breakeven analysis

- Multiple products
  - A single or constant mix of products is assumed.
  - Over time a firm may deviate from this and allocate its fixed costs in a different manner.
- Uncertainty
  - In price, fixed and variable cost
  - In reality, firms do their best to estimate/forecast what they think these will be when launching a product, and the breakeven quantity will only hold as long as estimates hold.
- Inconsistency in Planning Horizon
  - Benefits of some cost may take time to see, *e.g*. R&D may take more time to be realized in products than the typical breakeven analysis time horizon of one year.

Breakeven Analysis or Contribution Analysis?

- Breakeven analysis provides a good rule of thumb when determining how much output to produce to cover costs.
- A crucial problem with it, however, is that it ignores the fact that some costs are unavoidable. Therefore, those fixed costs are irrelevant in decision making.
- Contribution analysis considers this and instead asks if the incremental sales are enough to cover the increase in fixed costs associated with the change.

$$(P - VC) \times \Delta Q > \Delta \text{Total Fixed Cost}$$

- **Example**: Suppose a worker proposes a direct $500,000 payment (a fixed cost) to a local retailer for better/more shelf space. Your current product price is $200, variable cost is $125, and the estimated change in output is 5,000. Do you proceed with the proposed change?

$$(\$200 - \$125) \times 5000 > \$500{,}000$$

$$\$375{,}000 < \$500{,}000$$

Choosing Between Different Technologies

- Deciding between two technologies (or more) requires looking at the breakeven quantity between the two.
- Typically, a firm has a *high fixed/low marginal cost* technology or a *low fixed/high marginal cost* technology.
- If the firm expects to sell more than the breakeven quantity between the two technologies, then it is more beneficial to choose the lower marginal cost technology.
- Suppose we have **technology A** that has a fixed cost of \$800 with marginal costs of \$20, and **technology B** that has a fixed cost of \$100 with a marginal cost of \$70. What is the breakeven quantity between the two technologies?

$$\$800 + \$20Q = \$100 + \$70Q$$

$$\$700 = 50Q$$

$$Q = 14$$

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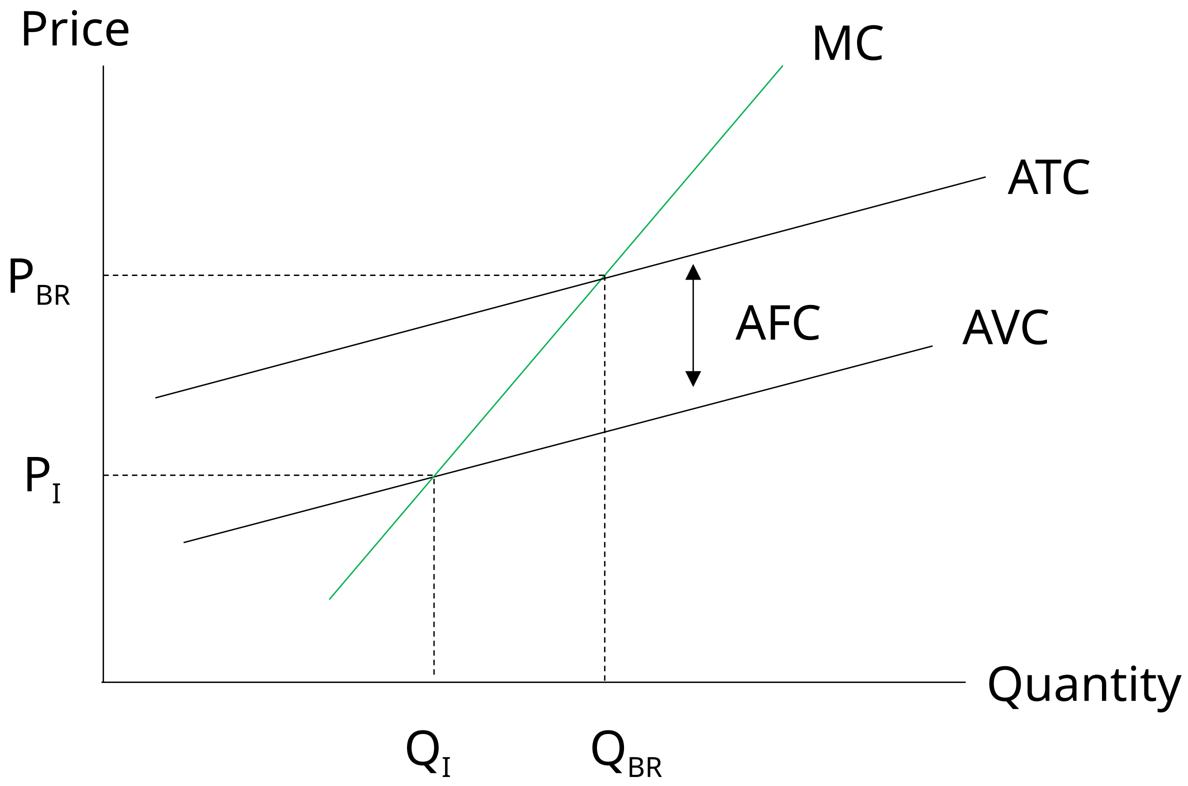
Shutting Down

- If a firm is faced with a price that is too low, such that it cannot cover its average variable cost then it is best for the firm to shut down.
- In this case, firms cannot cover the variable cost of production (wages, resources, etc.) much less its fixed cost of production.
- Graphically…

- Shutting down is different in the long run and short run. In the long run the firm adds in fixed costs into its calculation since fixed costs become avoidable.



Shutting Down Example

Price	Output	TR	MR	FC	VC	TC	MC	Profit/Loss	AFC	AVC	ATC
$20	80	$1,600		$300	$1,200	$1,500		$100	$3.75	$15.00	$18.75
19	90	$1,710	$110	$300	$1,300	$1,600	$100	$110	$3.33	$14.44	$17.78
18	100	$1,800	$90	$300	$1,390	$1,680	$80	$120	$3.00	$13.90	$16.90
17	110	$1,870	$70	$300	$1,460	$1,740	$60	$130	$2.73	$13.27	$16.00
16	**120**	**$1,920**	**$50**	**$300**	**$1,510**	**$1,790**	**$50**	**$130**	**$2.50**	**$12.58**	**$15.08**
15	130	$1,950	$30	$300	$1,710	$2,010	$220	($60)	$2.31	$13.15	$15.46
14	**140**	**$1,960**	**$10**	**$300**	**$2,150**	**$2,450**	**$440**	**($490)**	**$2.14**	**$15.36**	**$17.50**
13	150	$1,950	($10)	$300	$2,780	$3,080	$630	($1,130)	$2.00	$18.53	$20.53
12	160	$1,920	($30)	$300	$3,590	$3,890	$810	($1,970)	$1.88	$22.44	$24.31
11	170	$1,870	($50)	$300	$4,600	$4,900	$1,010	($3,030)	$1.76	$27.06	$28.82

Hold-Up

- Hold-up occurs when there are sunk costs, as one party can renege on an agreement.
- Suppose a firm has investment in a new technology (giving them a sunk cost).
  - The firm will use this technology to sell its output to another firm.
  - If the second firm reneges on the deal with the first firm, then we have a post-investment hold-up problem.
  - The hold-up occurs because we assume that the first firm cannot recoup any money on its technology (sunk cost) and there is no contract between the two firms.
- How do you eliminate the hold-up
  - Contracts are the simplest way.
  - The second firm can buy the technology then let the first firm lease it. (exchange of hostages)
    - Common in biotechnology/pharmaceuticals
  - Partnerships and mergers (vertical integration) offer another solution. Firms can buy up suppliers or patent holders (phone manufacturers)
  - If there is a threat of hold-up then an incentive (see above) has to be given to keep the relationship-specific investment going.
- Can we eliminate hold-up in dating/marriage?

Activity – Is It a Sunk Cost?

- Scenario A: You paid $50,000 last year for a machine. Today, it's worth $10,000 if sold.
- Scenario B: You just signed a 3-year lease for $6,000/month. You want to shut down after month 1.

Questions:

1. For each case, what part is sunk? What part is avoidable?
2. Should past spending influence today's decision?

Activity – Is It a Sunk Cost?

Scenario A: Equipment Purchase

- Sunk cost: $50,000 paid last year, it cannot be recovered.
- Relevant cost: $10,000 resale value, it’s an opportunity cost of keeping the machine.
- Decision insight: Ignore the $50K. Only compare keeping vs selling today based on current/future value.

Scenario B: Lease Contract

- If the lease is non-cancelable, then entire lease is sunk
- If you can terminate after a penalty, then only some portion is sunk; rest is avoidable.
- Decision insight: Consider only future avoidable costs, not past commitments.

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Discussion Activity 3

- Why do some companies merge with suppliers or R&D firms instead of contracting with them?
- Think about trust, bargaining power, and protecting relationship-specific investments.

Key Takeaways

- Investments involve trading present dollars for future gains, use tools to assess if it's worth it.
- Discount rate = opportunity cost of capital
  - Only invest when return > cost of capital.
- Net Present Value (NPV) is the gold standard
  - Accounts for time value, risk, and all cash flows.
- Payback period is simpler but ignores long-term profitability.
- Break-even analysis shows the quantity or price needed to cover costs.
  - Helps with pricing, shutdown, or expansion decisions.
- Contribution analysis focuses on whether a change adds more value than it costs.
- Sunk costs are irrelevant, don’t let past spending drive future choices.
- Post-investment hold-up can destroy value
  - Use contracts, ownership, or partnerships to protect specific investments.

Knowledge Check

1. Why is NPV a better decision-making tool than payback period?
2. How does the cost of capital affect investment decisions?
3. What is the break-even quantity if fixed costs = $1,000 and contribution margin = $10?
4. When should a firm shut down in the short run?
5. What is a post-investment hold-up and how can it be avoided?

Knowledge Check Answers

1. NPV accounts for all future cash flows and time value of money
2. Only invest when expected return > cost of capital
3. 1,000 / 10 = 100 units
4. When price < average variable cost
5. When one party reneges after another makes a sunk investment; avoided with contracts or integration

Consultant’s Challenge: Invest or Walk Away?

Scenario:

- You’re advising a boutique coffee chain considering opening two new locations. They’ve estimated high upfront costs, and they're unsure how to evaluate whether the investment makes sense.

Your Task:

1. How would you guide them in using tools like NPV, break-even analysis, or payback periods to decide?
2. What would you warn them about when it comes to sunk costs or misinterpreting the break-even point?

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